CORE CSE SERIES — MCQ EDITION

Computer Networks MCQ Handbook

Basic to Advanced Multiple-Choice Questions for Company OTs, PYQ-style Practice & Core Placement Preparation — with Answers & Explanations

OSI/TCP-IP | Data Link | Network Layer | Transport Layer | Application Layer | Security

Compiled for Product & Service-based Company Online Tests, GATE-style Core Practice, and Interview Rounds

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COMPUTER NETWORKS

Topic-wise MCQs with Explanations

1. OSI & TCP/IP Models Layers, encapsulation, protocol mapping

Q1 **How many layers does the OSI model have?** BASIC

A. 7
B. 4
C. 5
D. 6

Correct Answer: A. 7

Explanation: The OSI model has 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Q2 **Which OSI layer is responsible for routing packets between different networks?** BASIC

A. Network Layer
B. Data Link Layer
C. Transport Layer
D. Physical Layer

Correct Answer: A. Network Layer

Explanation: The Network layer (Layer 3) handles logical addressing (IP) and routing of packets across networks.

Q3 **The TCP/IP model condenses the OSI model into how many layers?** BASIC

A. 4
B. 7
C. 5
D. 3

Correct Answer: A. 4

Explanation: The TCP/IP model has 4 layers: Network Interface (Link), Internet, Transport, and Application — combining OSI's top three into one Application layer.

Q4 **Which OSI layer handles encryption and data compression?** INTERMEDIATE

A. Presentation Layer

B. Session Layer

C. Application Layer

D. Transport Layer

Correct Answer: A. Presentation Layer

Explanation: The Presentation layer handles data translation, encryption/decryption, and compression before passing data to the Application layer.

Q5 **Which layer is responsible for establishing, managing, and terminating sessions between applications?** INTERMEDIATE

A. Session Layer

B. Transport Layer

C. Network Layer

D. Application Layer

Correct Answer: A. Session Layer

Explanation: The Session layer manages dialog control — establishing, maintaining, and synchronizing communication sessions between two systems.

Q6 **Encapsulation in networking refers to:** INTERMEDIATE

A. Adding header (and sometimes trailer) information as data moves down the protocol stack

B. Compressing data for storage

C. Encrypting all network traffic

D. Converting analog to digital signals

Correct Answer: A. Adding header (and sometimes trailer) information as data moves down the protocol stack

Explanation: As data passes down each layer of the stack, that layer adds its own header (e.g., TCP header, then IP header, then Ethernet frame header) — this is encapsulation.

2. Physical & Data Link Layer Framing, error detection, MAC, switching

Q7 **What is the main function of the Data Link layer?** BASIC

A. Node-to-node delivery and framing of data, with MAC addressing

B. End-to-end delivery across networks

C. Routing between networks

D. Session management

Correct Answer: A. Node-to-node delivery and framing of data, with MAC addressing

Explanation: The Data Link layer handles framing, physical/MAC addressing, and reliable node-to-node (hop-to-hop) delivery within the same network segment.

Q8 **Which error-detection method uses polynomial division and is common in Ethernet frames?** INTERMEDIATE

A. CRC (Cyclic Redundancy Check)

B. Parity bit

C. Checksum only

D. Hamming code

Correct Answer: A. CRC (Cyclic Redundancy Check)

Explanation: CRC treats data as a polynomial and divides it by a generator polynomial; the remainder is appended as the frame check sequence, widely used in Ethernet.

Q9 **A MAC address is typically how many bits long?** BASIC

A. 48 bits

B. 32 bits

C. 64 bits

D. 16 bits

Correct Answer: A. 48 bits

Explanation: A standard MAC address is 48 bits (6 bytes), usually written as 12 hexadecimal digits, e.g., 00:1A:2B:3C:4D:5E.

Q10 **Which device operates primarily at the Data Link layer, forwarding frames based on MAC addresses?** BASIC

A. Switch
B. Router
C. Hub
D. Gateway

Correct Answer: A. Switch

Explanation: A switch learns MAC addresses and forwards frames only to the relevant port, unlike a hub which broadcasts to all ports.

Q11 **What is the purpose of the CSMA/CD protocol?** INTERMEDIATE

A. To detect and handle collisions on a shared Ethernet medium
B. To encrypt Ethernet frames
C. To assign IP addresses dynamically
D. To compress data before transmission

Correct Answer: A. To detect and handle collisions on a shared Ethernet medium

Explanation: Carrier Sense Multiple Access with Collision Detection lets stations sense the medium before transmitting and detect/handle collisions when they occur (used in legacy shared Ethernet).

Q12 **CSMA/CA (as used in Wi-Fi) differs from CSMA/CD primarily because:** ADVANCED

A. Wireless collisions are hard to detect, so CA focuses on collision avoidance instead of detection
B. CA does not need to sense the carrier at all
C. CA is used only in wired Ethernet
D. CA guarantees zero collisions

Correct Answer: A. Wireless collisions are hard to detect, so CA focuses on collision avoidance instead of detection

Explanation: In wireless networks, a station often can't reliably detect a collision while transmitting, so CSMA/CA uses techniques like RTS/CTS and random backoff to avoid collisions proactively.

Q13 **What is the purpose of the ARP (Address Resolution Protocol)?** INTERMEDIATE

A. To resolve/map an IP address to its corresponding MAC address on a local network
B. To resolve domain names to IP addresses
C. To assign IP addresses automatically
D. To route packets between networks

Correct Answer: A. To resolve/map an IP address to its corresponding MAC address on a local network

Explanation: ARP allows a device to find the MAC address associated with a known IP address on the same local network segment.

3. Network Layer — IP, Routing & Subnetting

IP addressing, subnetting, routing protocols

Q14 **How many bits are in an IPv4 address?** BASIC

A. 32 bits

B. 64 bits

C. 128 bits

D. 16 bits

Correct Answer: A. 32 bits

Explanation: IPv4 addresses are 32 bits long, typically represented as four decimal octets (e.g., 192.168.1.1).

Q15 **How many bits are in an IPv6 address?** BASIC

A. 128 bits

B. 32 bits

C. 64 bits

D. 256 bits

Correct Answer: A. 128 bits

Explanation: IPv6 addresses are 128 bits, vastly expanding the address space compared to IPv4's 32 bits.

Q16 **What is the default subnet mask for a Class C IP address?** BASIC

A. 255.255.255.0

B. 255.255.0.0

C. 255.0.0.0

D. 255.255.255.255

Correct Answer: A. 255.255.255.0

Explanation: Class C addresses use a default mask of 255.255.255.0 (/24), providing 256 addresses per network (254 usable hosts).

Q17 **Given the network 192.168.1.0/26, how many usable host addresses are available per subnet?** INTERMEDIATE

A. 62
B. 64
C. 30
D. 126

Correct Answer: A. 62

Explanation: /26 leaves 6 host bits → 2^6 = 64 total addresses; subtracting network and broadcast addresses gives 62 usable hosts.

Q18 **Which of these is a private (non-routable on the public internet) IP address range?** INTERMEDIATE

A. 10.0.0.0 – 10.255.255.255
B. 8.8.8.0 – 8.8.8.255
C. 172.32.0.0 – 172.32.255.255
D. 1.1.1.0 – 1.1.1.255

Correct Answer: A. 10.0.0.0 – 10.255.255.255

Explanation: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16 are the three reserved private IP ranges defined by RFC 1918.

Q19 **Which routing protocol is a 'distance-vector' protocol using hop count as its metric?** ADVANCED

A. RIP (Routing Information Protocol)
B. OSPF
C. BGP
D. IS-IS

Correct Answer: A. RIP (Routing Information Protocol)

Explanation: RIP is a classic distance-vector protocol that uses hop count (max 15) as its routing metric, exchanging full routing tables periodically.

Q20 **OSPF is classified as which type of routing protocol?** ADVANCED

A. Link-state
B. Distance-vector
C. Path-vector
D. Hybrid only

Correct Answer: A. Link-state

Explanation: OSPF (Open Shortest Path First) is a link-state protocol; each router builds a complete map of the network topology and computes shortest paths using Dijkstra's algorithm.

Q21 What is the purpose of NAT (Network Address Translation)? BASIC

A. To translate private IP addresses to a public IP address for internet access
B. To translate domain names to IP addresses
C. To encrypt IP packets
D. To assign MAC addresses

Correct Answer: A. To translate private IP addresses to a public IP address for internet access

Explanation: NAT allows multiple devices on a private network to share a single public IP address when communicating with the internet, translating addresses at the router/gateway.

Q22 What does TTL (Time To Live) in an IP packet header prevent? INTERMEDIATE

A. Packets from looping indefinitely in the network
B. IP address conflicts
C. Packet fragmentation
D. Duplicate MAC addresses

Correct Answer: A. Packets from looping indefinitely in the network

Explanation: TTL is decremented at each router hop; when it reaches zero, the packet is discarded, preventing infinite routing loops.

4. Transport Layer — TCP, UDP & Congestion Control

Reliability, flow control, congestion control, sockets

Q23 Which protocol establishes a connection using a 3-way handshake? BASIC

A. TCP
B. UDP
C. IP
D. ICMP

Correct Answer: A. TCP

Explanation: TCP uses the SYN, SYN-ACK, ACK three-way handshake to establish a reliable, connection-oriented session before data transfer.

Q24 **Which of these applications typically uses UDP rather than TCP?** BASIC

A. DNS queries and live video streaming
B. Web browsing (HTTP)
C. Email (SMTP)
D. File transfer (FTP)

Correct Answer: A. DNS queries and live video streaming

Explanation: UDP's low latency (no handshake, no retransmission delay) suits DNS lookups and real-time streaming/gaming where occasional loss is acceptable.

Q25 **What is the purpose of TCP's sliding window mechanism?** INTERMEDIATE

A. To perform flow control by regulating how much unacknowledged data can be in transit
B. To encrypt data in transit
C. To assign port numbers
D. To detect malware in packets

Correct Answer: A. To perform flow control by regulating how much unacknowledged data can be in transit

Explanation: The sliding window lets the receiver advertise how much buffer space it has, allowing the sender to send multiple segments before waiting for an ACK, improving throughput while preventing overload.

Q26 **Which TCP congestion control phase increases the congestion window exponentially?** ADVANCED

A. Slow Start
B. Congestion Avoidance
C. Fast Recovery
D. Fast Retransmit

Correct Answer: A. Slow Start

Explanation: During Slow Start, the congestion window doubles roughly every RTT until it hits a threshold (ssthresh) or a loss is detected.

Q27 **What triggers TCP's Congestion Avoidance phase to switch from Slow Start?** ADVANCED

A. When the congestion window reaches the slow start threshold (ssthresh)
B. When 3 duplicate ACKs are received
C. When the connection is first established
D. When the receiver's window is zero

Correct Answer: A. When the congestion window reaches the slow start threshold (ssthresh)

Explanation: Once cwnd reaches ssthresh, TCP switches from exponential growth (Slow Start) to linear (additive) growth in Congestion Avoidance.

Q28 A well-known port number range is: INTERMEDIATE

A. 0–1023
B. 1024–49151
C. 49152–65535
D. 0–65535 all equally well-known

Correct Answer: A. 0–1023

Explanation: Ports 0–1023 are reserved 'well-known' ports (e.g., HTTP=80, HTTPS=443, FTP=21), 1024–49151 are registered ports, and 49152–65535 are dynamic/private ports.

Q29 Which flag in a TCP segment is used to gracefully terminate a connection? INTERMEDIATE

A. FIN
B. SYN
C. RST
D. ACK only

Correct Answer: A. FIN

Explanation: The FIN flag signals that the sender has no more data to send, initiating a graceful connection termination (four-way handshake).

Q30 What does the RST flag in TCP indicate? ADVANCED

A. An abrupt/abnormal termination of the connection
B. A retransmission request
C. A request to resize the window
D. A request to start a new session

Correct Answer: A. An abrupt/abnormal termination of the connection

Explanation: RST (reset) is used to abruptly abort a connection, typically sent when a segment arrives for a connection that doesn't exist or after an unrecoverable error.

5. Application Layer Protocols HTTP, DNS, FTP, SMTP, DHCP

Q31 What is the default port number for HTTPS? BASIC

A. 443
B. 80
C. 21
D. 25

Correct Answer: A. 443

Explanation: HTTPS (HTTP over TLS/SSL) uses port 443 by default, while HTTP uses port 80.

Q32 **Which protocol is used to send emails from a client to a mail server?** BASIC

A. SMTP

B. POP3

C. IMAP

D. FTP

Correct Answer: A. SMTP

Explanation: SMTP (Simple Mail Transfer Protocol) is used to send/relay outgoing mail; POP3/IMAP are used by clients to retrieve/read received mail.

Q33 **What is the difference between POP3 and IMAP?** INTERMEDIATE

A. POP3 typically downloads and removes mail from the server; IMAP keeps mail synced on the server across devices

B. POP3 is used for sending mail; IMAP is used for browsing the web

C. POP3 works only over UDP; IMAP works only over TCP

D. There is no difference

Correct Answer: A. POP3 typically downloads and removes mail from the server; IMAP keeps mail synced on the server across devices

Explanation: POP3 traditionally downloads mail locally (often deleting from server), while IMAP keeps messages on the server, allowing multi-device sync.

Q34 **DHCP is primarily used to:** BASIC

A. Automatically assign IP addresses and network configuration to devices

B. Encrypt network traffic

C. Translate domain names to IP addresses

D. Route packets between networks

Correct Answer: A. Automatically assign IP addresses and network configuration to devices

Explanation: The Dynamic Host Configuration Protocol automatically assigns IP addresses, subnet masks, gateways, and DNS servers to devices joining a network.

Q35 **Which HTTP method is idempotent and used to retrieve a resource?** BASIC

A. GET

B. POST

C. PATCH

D. CONNECT

Correct Answer: A. GET

Explanation: GET requests are idempotent (repeating them has the same effect) and are used to retrieve resource representations without side effects.

Q36 **HTTP status code 404 indicates:** BASIC

A. Resource Not Found
B. Server Error
C. Unauthorized
D. Redirection

Correct Answer: A. Resource Not Found

Explanation: 404 Not Found means the server couldn't find the requested resource; 5xx codes indicate server errors, 401/403 indicate auth issues, 3xx indicate redirection.

Q37 **What does a DNS 'CNAME' record do?** INTERMEDIATE

A. Maps an alias domain name to another canonical domain name
B. Maps a domain name directly to an IPv4 address
C. Maps a domain name to a mail server
D. Stores text metadata for a domain

Correct Answer: A. Maps an alias domain name to another canonical domain name

Explanation: A CNAME (Canonical Name) record aliases one domain to another, which is then resolved further (an A record eventually resolves to an IP).

Q38 **An 'A record' in DNS maps a domain name to:** BASIC

A. An IPv4 address
B. An IPv6 address
C. A mail server
D. Another domain name

Correct Answer: A. An IPv4 address

Explanation: An A record maps a hostname directly to an IPv4 address; the equivalent for IPv6 is an AAAA record.

Q39 **FTP typically uses which two ports for control and data connections respectively?** INTERMEDIATE

A. 21 and 20
B. 80 and 443
C. 25 and 110
D. 53 and 67

Correct Answer: A. 21 and 20

Explanation: FTP uses port 21 for control commands and port 20 for the actual data transfer in active mode.

6. Network Security & Advanced Topics

Firewalls, VPN, load balancing, wireless, misc advanced

Q40 **What is the primary purpose of a firewall?** BASIC

A. To monitor and filter incoming/outgoing network traffic based on security rules

B. To speed up internet connections

C. To assign IP addresses

D. To cache web pages

Correct Answer: A. To monitor and filter incoming/outgoing network traffic based on security rules

Explanation: A firewall enforces a security policy by allowing or blocking traffic based on rules (IP, port, protocol, etc.), forming a barrier between trusted and untrusted networks.

Q41 **A VPN (Virtual Private Network) primarily provides:** BASIC

A. A secure, encrypted tunnel over a public network to access a private network remotely

B. Faster internet speeds always

C. Free public Wi-Fi

D. Automatic malware removal

Correct Answer: A. A secure, encrypted tunnel over a public network to access a private network remotely

Explanation: VPNs create an encrypted tunnel between a client and a remote network, ensuring confidentiality and secure access as if directly connected to the private network.

Q42 **What is a 'man-in-the-middle' attack?** INTERMEDIATE

A. An attacker secretly intercepts and possibly alters communication between two parties

B. A brute-force password attack

C. A denial-of-service flood attack

D. A phishing email attack only

Correct Answer: A. An attacker secretly intercepts and possibly alters communication between two parties

Explanation: In a MITM attack, the attacker positions themselves between two communicating parties to eavesdrop on or manipulate the exchanged data without either party's knowledge.

Q43 **What is the main goal of a Denial-of-Service (DoS) attack?** BASIC

A. To make a service/resource unavailable to legitimate users by overwhelming it
B. To steal encrypted data
C. To intercept private keys
D. To modify DNS records

Correct Answer: A. To make a service/resource unavailable to legitimate users by overwhelming it

Explanation: DoS (and distributed DoS/DDoS) attacks flood a target with traffic/requests to exhaust its resources, denying service to legitimate users.

Q44 **What does SSL/TLS primarily provide in a network connection?** INTERMEDIATE

A. Encryption, authentication, and data integrity
B. Faster routing
C. IP address translation
D. Load balancing

Correct Answer: A. Encryption, authentication, and data integrity

Explanation: SSL/TLS encrypts data in transit, authenticates the server (and optionally the client) via certificates, and ensures data hasn't been tampered with.

Q45 **In a CDN (Content Delivery Network), content is typically:** INTERMEDIATE

A. Cached at geographically distributed edge servers closer to users to reduce latency
B. Stored only on a single central server
C. Encrypted using client-side keys only
D. Compressed but never cached

Correct Answer: A. Cached at geographically distributed edge servers closer to users to reduce latency

Explanation: CDNs replicate/cache content across globally distributed edge servers so users are served from a nearby node, reducing latency and origin server load.

Q46 **Which load balancing algorithm routes based on which server currently has the fewest active connections?** ADVANCED

A. Least Connections
B. Round Robin
C. IP Hash
D. Random

Correct Answer: A. Least Connections

Explanation: The Least Connections algorithm dynamically routes new requests to the server with the fewest currently active connections, useful when requests have variable durations.

Q47 **What is 'subnetting' primarily used for?** BASIC

A. Dividing a large network into smaller, more manageable sub-networks

B. Encrypting IP packets

C. Assigning MAC addresses

D. Compressing IP headers

Correct Answer: A. Dividing a large network into smaller, more manageable sub-networks

Explanation: Subnetting splits a network address space into smaller subnets, improving address utilization, security segmentation, and routing efficiency.

Q48 **What is the purpose of the 3-way handshake's final ACK in TCP connection establishment?** INTERMEDIATE

A. To confirm both sides are ready and synchronized before data transfer begins

B. To close the connection

C. To request retransmission

D. To assign a new port number

Correct Answer: A. To confirm both sides are ready and synchronized before data transfer begins

Explanation: After SYN and SYN-ACK, the final ACK confirms receipt and synchronization of sequence numbers from both sides, completing connection establishment.